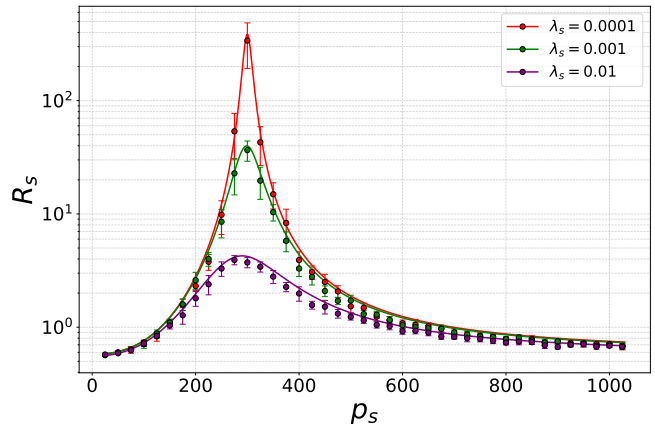
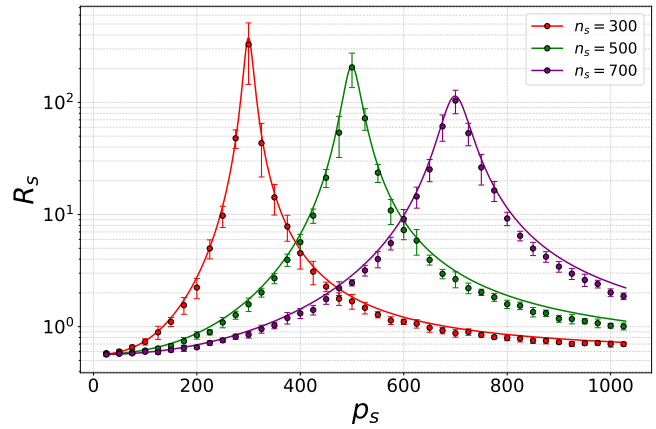
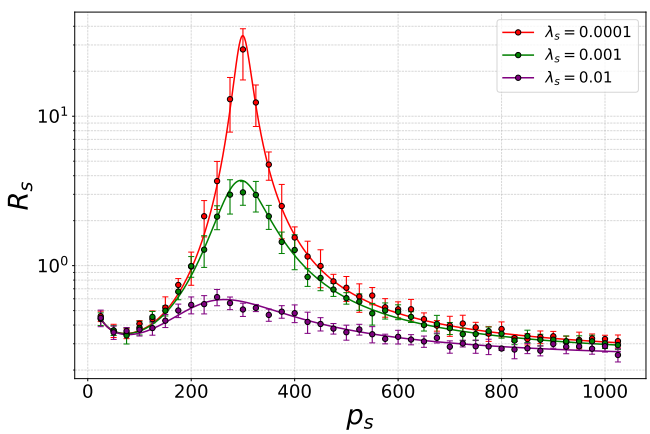


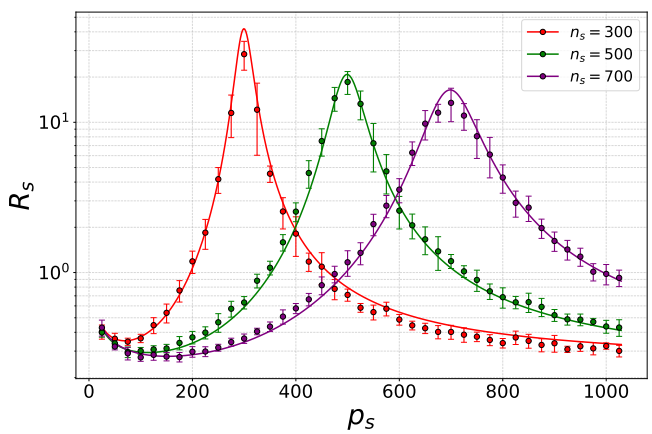
(a) Single-index target function, $n_\mathsf{s} = 300$.



(b) Single-index target function, $\lambda_\mathsf{s} = 10^{-4}$.

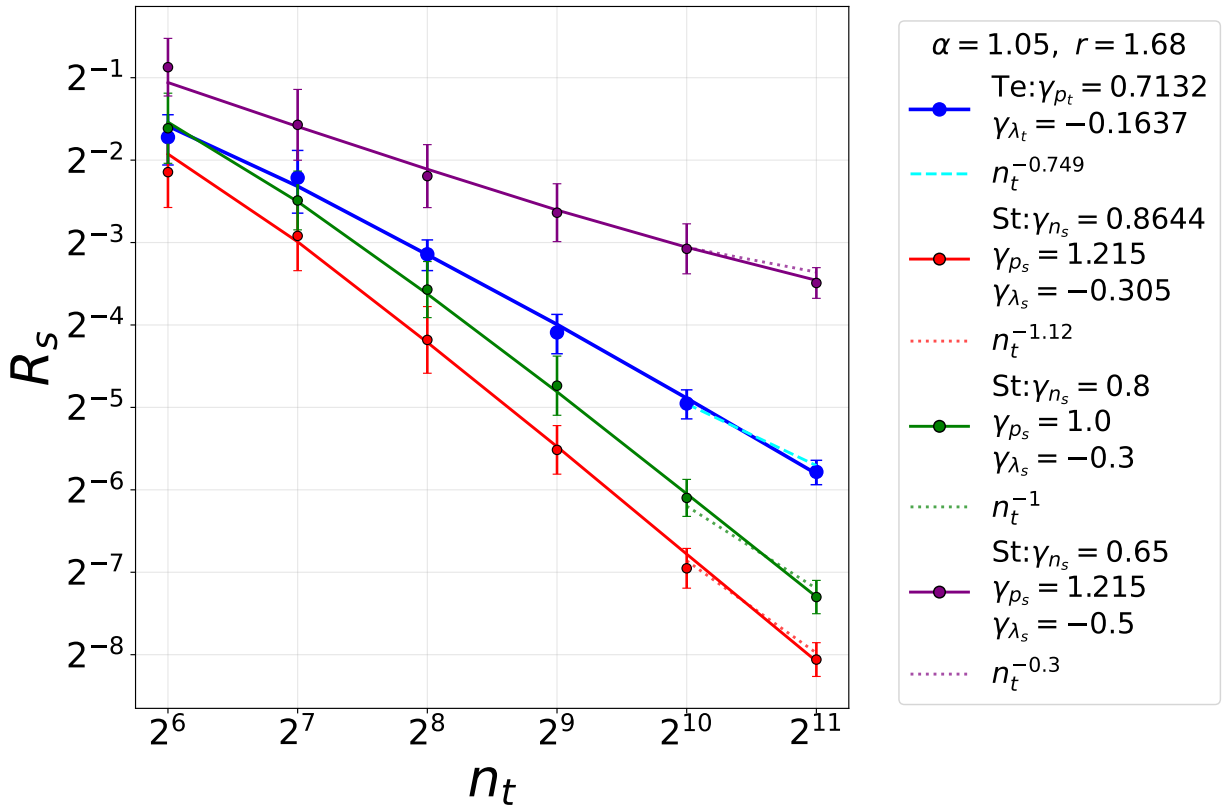


(c) MNIST data, $n_\mathsf{s} = 300$.



(d) MNIST data, $\lambda_\mathsf{s} = 10^{-4}$.

Figure 1: Excess test errors of various students trained on teacher labels, together with the corresponding deterministic equivalents (Theorem 2), as a function of the number of student features p_s. The teacher and the student are random feature models with the same number of features ($p_\mathsf{t} = p_\mathsf{s}$), trained with the same sample size ($n_\mathsf{t} = n_\mathsf{s}$) and with the teacher regularization being half of that of the student ($\lambda_\mathsf{t} = \lambda_\mathsf{s}/2$). In the top two plots, the data is given by $\boldsymbol{x}_i \sim \mathcal{N}(0, I_d)$ and $y_i = \mathrm{erf}(\langle \boldsymbol{x}_i, \boldsymbol{w}_* \rangle) + \varepsilon_i$, with $\boldsymbol{w}_* \sim \mathrm{Unif}(\mathbb{S}^{d-1})$, and the random feature model is $\varphi(\boldsymbol{x}; \boldsymbol{w}) = \tanh(\langle \boldsymbol{x}, \boldsymbol{w} \rangle)$, $\boldsymbol{w} \sim \mathrm{Unif}(\mathbb{S}^{d-1})$; in the bottom two plots, the data is obtained from the MNIST dataset and the random feature model is $\varphi(\boldsymbol{x}; \boldsymbol{w}) = \mathrm{erf}(\langle \boldsymbol{x}, \boldsymbol{w} \rangle)$, $\boldsymbol{w} \sim \mathrm{Unif}(\mathbb{S}^{d-1})$. In the left figures, we fix the sample size n_s and consider three values of the regularization λ_s (in three different colors); in the right figures, we fix the regularization λ_s and consider three values of the sample size n_s (in three different colors). We run 10 independent experiments, reporting the average and the confidence interval at 1 standard deviation. The circles represent the test errors obtained experimentally and they match well the continuous curves which represent the deterministic equivalents of Definition 2.



(a) Bias dominated: W2SG.

Figure 2: Excess test errors of various students and teachers, as a function of the number of ground-truth samples n_{t}. We consider random feature ridge regression with weak-to-strong training, under the source and capacity conditions in (16) and with the hyperparameter scaling in (17). Points are empirical experiments, solid lines are deterministic equivalent predictions (Theorems 1 and 2), dotted lines are theoretical decay rates (Theorems 3 and 4). The teacher test error is in blue, while the other colors correspond to student test errors for different choices of $(\gamma_{n_s}, \gamma_{p_s}, \gamma_{\lambda_s})$.